

# YASH ARORA

AI / ML Engineer · Applied AI & Generative AI · Full-Stack Delivery

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AWS ML pipeline · 92–96% CV accuracy · ~30% RAG hallucination cut · Full MLOps + CI/CD · 1,500+ learners enabled

## SUMMARY

B.Tech AI (2025) who ships applied ML and Generative AI systems end-to-end — train, evaluate, containerize, deploy. Production signals: AWS-deployed scoring service with MLflow + DVC + GitHub Actions; hybrid BM25 + dense RAG pipeline that cut unsupported answers ~30% on a held-out set; CV models at 92–96% accuracy served via FastAPI/Flask. Hands-on with LangChain, vector DBs, LLMs, model evaluation, and explaining model risk & rollout to non-technical stakeholders. Seeking ML Engineer / Applied AI / GenAI / AI Consultant roles in Delhi NCR.

## TECHNICAL SKILLS

**ML / DL:** Python, scikit-learn, XGBoost, CatBoost, TensorFlow, PyTorch, OpenCV, YOLOv5, Hugging Face, NumPy, Pandas

**LLMs & GenAI:** LangChain, RAG pipelines, hybrid retrieval (BM25 + dense), ChromaDB, Pinecone, OpenAI API, LLMs, Prompt Engineering, CrewAI

**MLOps & Cloud:** Docker, docker-compose, AWS (EC2, Elastic Beanstalk, ECR, SageMaker), MLflow, DVC, DagsHub, GitHub Actions CI/CD

**Software:** FastAPI, Flask, Django, REST APIs, React, SQL, MySQL, Git · **Languages:** Python, JavaScript, SQL, C++, Bash

## PROJECTS

### Student Exam Performance Pipeline — Deployed on AWS · 2026

- End-to-end tabular ML pipeline (scikit-learn / XGBoost / CatBoost) with MLflow experiment tracking and DVC + DagsHub versioning.
- Containerized with Docker; GitHub Actions builds and deploys the Flask scoring service to AWS on every push to main. Full ownership from training to production endpoint.

Tech: Python, scikit-learn, XGBoost, CatBoost, Flask, Docker, AWS, MLflow, DVC, GitHub Actions

### RAG Chatbot with Hybrid Retrieval · May 2024

- Crawler → chunker → ChromaDB → GPT-4. Hybrid BM25 + dense embeddings so keyword and semantic queries both retrieve.
- Held-out 200-question evaluation set: ~30% fewer unsupported answers vs. the same model with no retrieval (method documented in repo).

Tech: Python, LangChain, ChromaDB, OpenAI embeddings, BeautifulSoup

### Potato Disease Classifier (Served Full-Stack) · Mar 2024

- ResNet on PlantVillage (10,000+ images + augmentation): 96.3% test accuracy across 3 classes (+4 pp vs VGG16 baseline).
- FastAPI inference in Docker + React upload UI; entire stack starts with one docker-compose up. Production-ready local/cloud packaging.

Tech: TensorFlow, ResNet, FastAPI, Docker, React

### Real-time Drowsiness Detector · Aug 2024

- Fine-tuned YOLOv5 on 5,000 custom images annotated via Roboflow: 94% detection accuracy at <100 ms inference on laptop CPU.
- Deployed as Flask web app with live webcam feed; audio alerts trigger within 2 consecutive frames of eye closure.

Tech: YOLOv5, OpenCV, Flask, Transfer Learning, Python

## EXPERIENCE

### AI/ML Education Specialist · Pyxis Blu, Ranchi

Jan 2026 – Present

- Designed and shipped K–12 AI & robotics curriculum now live in 5+ partner schools; structured handbooks and rubrics cut teacher onboarding time ~50%.
- Built internal training portal (recordings, quizzes, progress tracking) used by 20+ teachers; programme reach scaled 3× beyond in-person sessions — full-stack product ownership.

### AI Trainer & Robotics Engineer (LMS Owner) · Reboot Robotics Academy, Ranchi

Nov 2024 – Mar 2025

- Designed and shipped a full-stack LMS from scratch (Django + MySQL + Bootstrap): course content, student records, progress tracking. Cut administrative overhead ~60% by replacing spreadsheet workflows with auth, CRUD, and reporting.
- Led 8 applied-ML bootcamps; 30+ learners shipped a first deployed model. Contributed to 600+ sessions across 20+ schools reaching 1,500+ students.

### AI/ML Intern · YHills Ed-Tech · Remote

Feb – Mar 2024

- Trained CNN/ANN image models (TensorFlow, scikit-learn) to 92% test accuracy; automated augmentation pipelines on 10,000+ images, cutting training time ~25%.
- Delivered weekly model performance dashboards (Matplotlib, Pandas) that surfaced data drift early and enabled 2× faster iteration for the research team.

### AI Model Evaluator (Freelance) · Scale AI · Remote

Oct 2023 – Feb 2024

- Reviewed 500+ model outputs across code generation, reasoning, and safety tasks; rated responses and flagged policy violations.
- Developed evaluation rubrics later adopted by the broader annotation team; contributed to content-safety pipelines across 1,000+ tasks — direct experience with model risk and evaluation design.

### ML Associate Executive · Google Developer Student Club (GDSC)

Aug 2023 – Aug 2024

- Hosted 10+ ML/GenAI webinars (Transformers, LLMs) for 300+ attendees; co-organized inter-college AI hackathon (120+ participants) as lead technical judge. Mentored 50+ developers — 15+ secured first AI/data internships.

## EDUCATION

**B.Tech, Artificial Intelligence** — Sarala Birla University, Ranchi · Apr 2021 – May 2025 · SGPA 7.25

Coursework: Deep Learning, Natural Language Processing (NLP), Computer Vision, Data Structures, Cloud Computing

## CERTIFICATIONS

- Machine Learning Specialization — Stanford / DeepLearning.AI (2023) · Multi-AI Agent Systems with CrewAI — DeepLearning.AI (2024)
- Machine Learning with Python — IBM / Coursera (2023) · Meta Front-End Developer Professional Certificate — Meta / Coursera (2024)